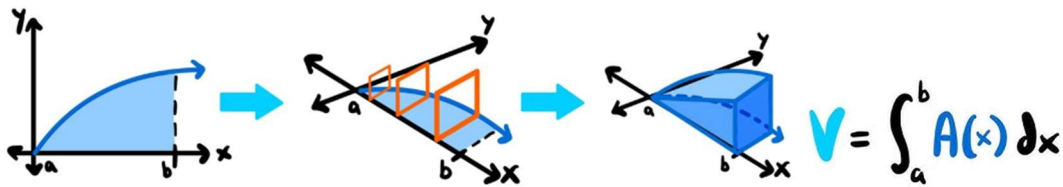
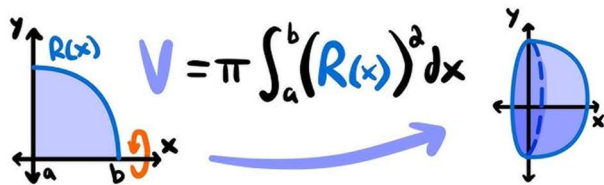


VOLUME OF KNOWN CROSS SECTIONS



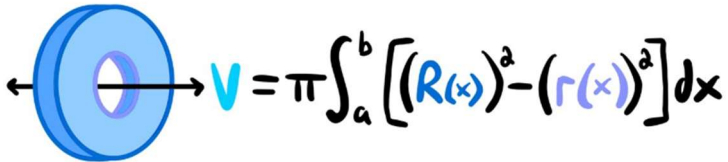
Area modeled by the function $A(x)$

DISK METHOD



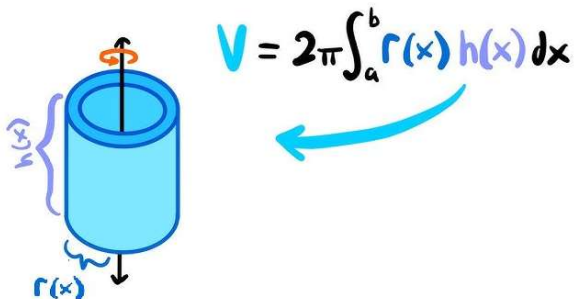
The Radius could also be $R(x) = f(x) - c$, the function revolving around an axis $y = c$, which would be different than the x axis

WASHER METHOD



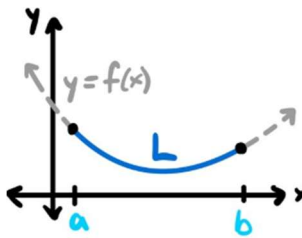
The Radii could also be $R(x) = f(x) - c$, $r(x) = g(x) - c$, if both are revolving around an axis $y = c$, which would be different than the x axis

SHELL METHOD



Here, the height of the cylinder is the difference between the two functions, $h(x) = f(x) - g(x)$, while the radius would be: $r(x) = x - c$, in case the axis of rotation is not the origin (in which case $r(x) = x$), but an axis $x = c$

ARC LENGTH



A graph showing a curve $y = f(x)$ on a Cartesian coordinate system. The curve is blue and lies above the x-axis. The x-axis is marked with points a and b . The arc length of the curve from $x = a$ to $x = b$ is labeled L .

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

AREA OF SURFACES OF REVOLUTION



A diagram showing a blue surface of revolution S generated by rotating a curve $y = f(x)$ around the x-axis. The surface is a horn-like shape. The x-axis is marked with points a and b .

$$S = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx$$

Surface Area of Revolution Formulas

The choice of formula depends on the axis of rotation and the variable of integration. ☺

- **Rotation about the x -axis:**

- If $y = f(x)$ on $[a, b]$: $S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

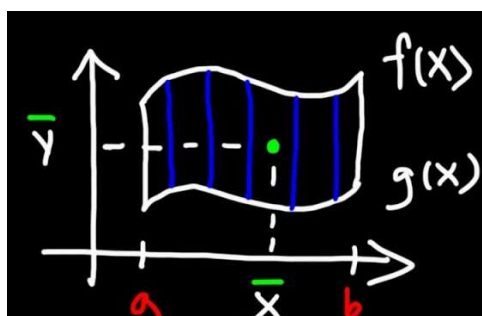
- If $x = h(y)$ on $[c, d]$: $S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

- **Rotation about the y -axis:**

- If $y = f(x)$ on $[a, b]$: $S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

- If $x = h(y)$ on $[c, d]$: $S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

CENTROIDS



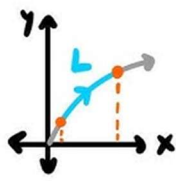
A diagram showing a region bounded by two curves $f(x)$ and $g(x)$ on the interval $[a, b]$. The region is shaded with vertical blue lines. The centroid is marked with a green dot at $(\bar{x}, \bar{y})$. The x-axis is marked with points a and b , and the y-axis is marked with $\bar{y}$.

$$A = \int_a^b [f(x) - g(x)] dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{1}{2} [f^2(x) - g^2(x)] dx$$

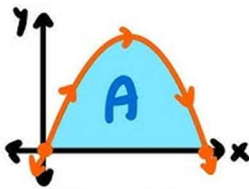
ARC LENGTH OF PARAMETRIC EQUATIONS



A diagram showing a coordinate system with x and y axes. A blue curve starts at the origin and goes up and to the right. A dashed vertical line drops from a point on the curve to the x-axis. The arc length of the curve from the origin to that point is labeled with a blue 'L'.

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

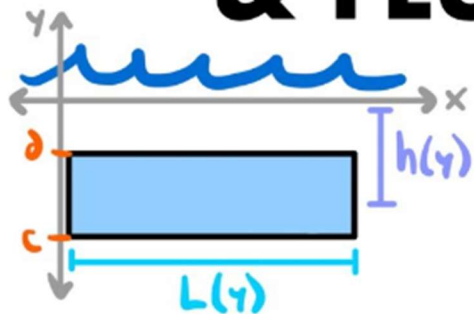
AREA UNDER PARAMETRIC EQUATIONS



A diagram showing a coordinate system with x and y axes. An orange curve starts at the origin, goes up and to the right, and then comes back down to the x-axis. The area under the curve is shaded in light blue and labeled with a blue 'A'.

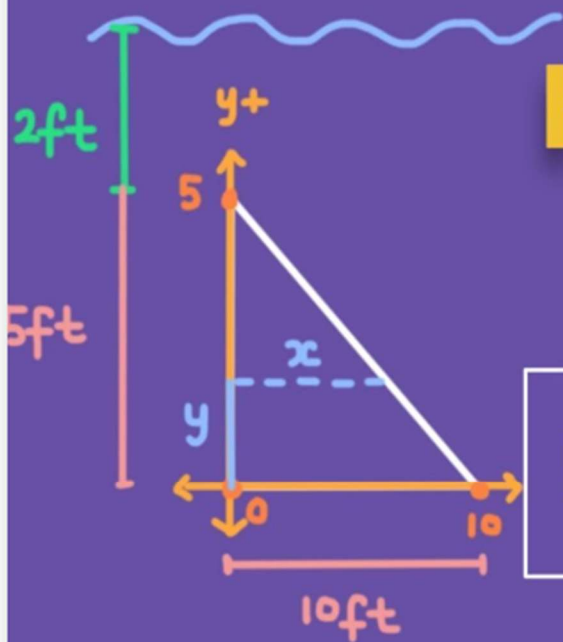
$$A = \int_{t_1}^{t_2} g(t) f'(t) dt$$

FLUID PRESSURE & FLUID FORCE



$$F = w \int_c^d h(y) L(y) dy$$

Hydrostatic Force

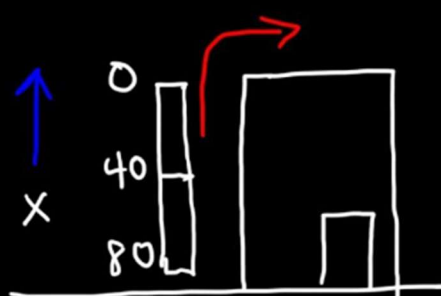


$$F = \int_a^b \text{density} \cdot \text{length} \cdot \text{depth} dy$$

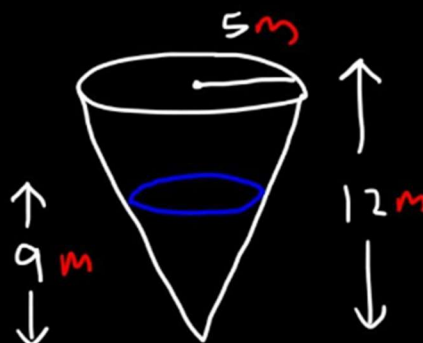
$$x = \frac{10}{5} = \frac{x}{5-y}$$

$$F = \int_0^5 62.5 \cdot (10-2y) \cdot (7-y) dy$$

Work Problems



$$W = \int_a^b F(x) dx$$



$$W = \rho g \int_a^b V(x) \Delta x dx$$

Ex.

A cable that weighs 3 pounds per foot is used to lift 1000 pounds of iron ore up a mineshaft which is 300 feet deep. Find the work done.

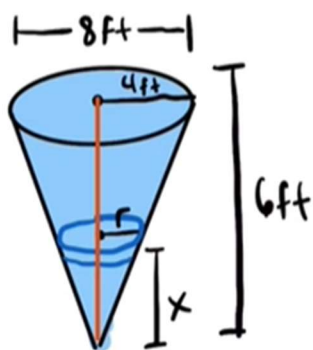


$$F(x) = \frac{lb}{ft} x = 3x + 1000$$

$$W = \int_a^b F(x) dx$$

Ex.

An open tank filled with water has the shape of a right circular cone. The tank is 8 feet across the top and 6 feet high. How much work is done in emptying the tank by pumping the water over the top edge?



$$A(x) = \pi r^2$$

$$= \pi \left(\frac{2}{3}x\right)^2$$

$$\Delta x = 6 - x$$

$$W = \rho g \int_a^b A(x) \Delta x dx$$

$$= 62.4 \int_0^6 \pi \left(\frac{2}{3}x\right)^2 (6-x) dx$$